



Aggregate data for analysis

Welcome back! In the next few videos, we'll explore something called data aggregation.

Aggregation means collecting or gathering many separate pieces into a whole. For example, the Milky Way galaxy is an aggregation of stars, dust, and gases. So data aggregation is the process of gathering data from multiple sources in order to combine it into a single summarized collection. In data analytics, a summarized collection, or summary, describes identifying the data you need and gathering it all together in one place. For example, let's say you have a cabinet full of different puzzles.

One day, a shelf breaks, and all the boxes topple over, scattering the puzzle pieces everywhere. To get each puzzle organized again, you need to identify the pieces that correspond to each particular puzzle, gather them together and put them back into their correct boxes. Only then can you work with these pieces and create a complete picture. So in data, the puzzle pieces represent the data that lives in different, separate datasets. Getting them organized is the aggregation process. Then the piles of pieces that complete a single puzzle become your summary. And finally, putting those pieces back together is like analyzing them to gain important insights.

Data aggregation helps data analyst identify trends, make comparisons and gain insights that wouldn't be possible if each of the data elements were analyzed on its own.

For instance, data on high school graduations for individual students can be aggregated

into a single graduation rate for an entire class. Data can also be aggregated over a given time period to provide statistics, such as averages, minimums, maximums, and sums. For example, that same yearly graduation rate data can be aggregated once again into a summary that shows us graduation rates for districts, states, and countries. Here's another example. Let's say you had data on real estate sales in a particular neighborhood for each of the past 10 years. If you aggregated all of that data, you'd be able to discover the average price of a home in that area and how values have increased or decreased over time.

Functions are a big help in making data aggregation possible. You'll learn how to use some of the most common ones to create your summaries soon. In addition, we'll talk about aggregating data using something called a subquery. You've seen SQL in action, and you understand that a query is a request for information from a database. So a subquery, also called an inner or nested query, is a query within another query. After the next several videos, you'll know how to aggregate data and understand the tools you'll be using along the way. Let's get started!